

API Security Risk Analysis Report

Task: Task 3 – API Security Risk Analysis

1. Introduction

This report analyzes the security posture of the public JSONPlaceholder API using Postman. The objective is to identify common API security risks and recommend improvements.

2. Tools Used

- Postman
- Browser
- JSONPlaceholder Public API
- GitHub (for documentation)

3. API Tested

<https://jsonplaceholder.typicode.com/users>

4. Test Results

Issue	Observation	Risk	Recommendation
Missing Authentication	API accessible without login, API key or token.	Medium	Implement JWT, OAuth2 or API Keys.
Sensitive Data Exposure	Response includes email, phone, address, website and company information.	Medium	Return only necessary fields and protect sensitive information.
No Rate Limiting	Repeated requests were accepted without restriction.	Medium	Implement rate limiting and throttling.
Input Validation	Invalid IDs (/users/999999) and invalid input (/users/abc) returned 404 Not Found.	Low	Continue validating input and return safe error messages.

5. Conclusion

The API correctly handles invalid resource requests by returning 404 Not Found. Improvements are recommended in authentication, exposure of sensitive information, and rate limiting to strengthen API security.

6. Screenshots

Insert your Postman screenshots here before converting to PDF.